

Top 100 Data Analyst Interview Questions

Mastering the Essentials of Data Analytics (DA) & Business Intelligence (BI)

Section 1: SQL Fundamentals

1. What is a window function?
2. What is the difference between RANK() and ROW_NUMBER()?
3. How do you find the second highest salary?
4. What is a recursive CTE?
5. What is the difference between correlated and non-correlated subquery?
6. How do you remove duplicates without DISTINCT?
7. What is an INDEX and when do you use it?
8. Explain self-join with example.



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9. What is the difference between DELETE, DROP, and TRUNCATE?
10. How do you pivot/unpivot data in SQL?
11. What is LAG() and LEAD()?
12. How do you handle NULL in aggregates?
13. What is the difference between VIEW and MATERIALIZED VIEW?
14. Explain ACID properties.
15. How do you optimize a slow query?
16. What is the difference between INNER JOIN and EXISTS?
17. What is a FULL OUTER JOIN?
18. How do you find duplicates across tables?
19. What are SQL constraints?
20. Explain GROUPING SETS.



Section 2: Python Fundamentals

21. How do you handle missing data in Pandas?
22. What is the difference between `loc[]` and `iloc[]`?
23. What are lambda functions in data analysis?
24. How do you remove duplicates from DataFrame?
25. Explain `groupby()` and `agg()`.
26. How do you merge/join DataFrames?
27. What is vectorization?
28. How do you handle outliers using IQR method?
29. What is the difference between list, tuple, dict?
30. How do you pivot data with `pivot_table()`?



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- 31. What libraries do you use for viz (Matplotlib/Seaborn)?
- 32. Explain apply() vs map() vs applymap().
- 33. How do you read CSV with chunks?
- 34. What is NumPy broadcasting?
- 35. How do you handle time-series with Pandas (resample, shift)?
- 36. How do you calculate correlation matrix?
- 37. How do you optimize Pandas performance (e.g., dtype)?
- 38. What is the difference between Series vs DataFrame?
- 39. How do you filter DataFrame conditionally?
- 40. How do you rename columns efficiently?



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Section 3: Power BI Fundamentals

- 41. What is DAX?
- 42. What is the difference between Power Query and Power Pivot?
- 43. What is the difference between measure vs calculated column?
- 44. Explain CALCULATE() function.
- 45. What are relationships (1:1, 1:M, M:M)?
- 46. How do you handle many-to-many?
- 47. What is row-level security (RLS)?
- 48. How do you setup incremental refresh?
- 49. What is the difference between filters vs slicers?
- 50. What is a data model?



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- 51. How do you publish and share reports?**
- 52. What is performance analyzer tool?**
- 53. How do you create month-on-month growth DAX?**
- 54. How do you use custom visuals?**
- 55. What is gateway for refresh?**
- 56. What is a .pbix file?**
- 57. What are quick measures examples?**
- 58. What is data blending in Power BI?**
- 59. How do you optimize large datasets?**
- 60. What is Get Data?**



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Section 4: Excel Fundamentals

- 61. What is INDEX-MATCH combo?
- 62. How do you use Power Query?
- 63. What are dynamic arrays (FILTER, UNIQUE)?
- 64. What is SUMIFS vs SUMPRODUCT?
- 65. What is Power Pivot?
- 66. What are Goal Seek and Solver?
- 67. What are conditional formatting rules?
- 68. How do you remove duplicates advanced?
- 69. How do you create sparklines?
- 70. How do you use INDIRECT function?



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- 71. What are data validation lists?**
- 72. What are array formulas (Ctrl+Shift+Enter)?**
- 73. What are Macros/VBA basics?**
- 74. What are pivot table slicers?**
- 75. What is what-if analysis?**
- 76. What are TEXTJOIN and CONCAT?**
- 77. What are XLOOKUP advantages?**
- 78. What is Flash Fill feature?**
- 79. What are Table vs Range pros?**
- 80. How do you import from SQL/Web?**



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Section 5: Tableau Fundamentals

- 81. What is data blending?
- 82. What is the difference between extract and live?
- 83. What are LOD expressions (FIXED, INCLUDE)?
- 84. What is the difference between parameters vs filters?
- 85. What is a blended axis?
- 86. What are Tableau Public features?
- 87. How do you setup cascading filters?
- 88. How do you do performance optimization?
- 89. What is published vs embedded data source?
- 90. How does Tableau generate SQL?



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- 91. What is Level of Detail calc?**
- 92. What are actions (filter, highlight)?**
- 93. What are dashboard extensions?**
- 94. What is row-level security?**
- 95. What is custom SQL connection?**
- 96. What are sets and groups?**
- 97. What are dual-axis charts?**
- 98. What are trend lines/stats?**
- 99. What is data densification?**
- 100. What are server administration basics?**



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